

Intelligent Public Health Surveillance and Disease Outbreak Prediction System

Course Code: CSA1013 | Course Title: Software Engineering

CO2 Assessment Tool 1 – Scenario-Based Requirement Analysis

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SOFTWARE REQUIREMENT SPECIFICATION (SRS)

1. Introduction

1.1 Purpose

This Software Requirement Specification (SRS) document provides a complete and detailed description of the functional and non-functional requirements for the Intelligent Public Health Surveillance and Disease Outbreak Prediction System. It is intended to serve as the single authoritative reference for the development, testing, and evaluation of the proposed system, and to communicate the expected behaviour of the system to all stakeholders, including project sponsors, software architects, developers, quality assurance teams, and public health domain experts.

The document translates the informal problem scenario provided by the public health agency into a structured, verifiable set of requirements that can be used to design system architecture, plan development sprints, and validate the final product against stakeholder expectations. It also documents the assumptions, constraints, and open questions identified during the analysis phase so that they can be resolved before implementation begins.

1.2 Document Conventions

Functional requirements are identified using the prefix "FR" followed by a sequential number (e.g., FR1, FR2). Non-functional requirements are grouped by quality attribute (Performance, Security, Reliability, etc.). Mandatory requirements use the keyword "shall"; desirable, non-mandatory features use the keyword "should". Actor names and use case names are capitalized for clarity throughout the document.

1.3 Intended Audience and Reading Suggestions

This document is intended for the following readers, each of whom may focus on different sections:

- Project Sponsors / Public Health Officials — Sections 1, 2, and 9 provide a high-level overview of objectives and system features.
- Software Architects and Developers — Sections 3, 4, 5, 7, and 8 provide the technical detail required for design and implementation.
- Quality Assurance / Testing Teams — Sections 4, 5, and 6 provide testable requirements and use case flows suitable for deriving test cases.
- Domain Experts / Epidemiologists — Sections 2, 6, and 10 describe the problem context, actor interactions, and operational assumptions.

1.4 Business Domain

The business domain addressed by this system is Public Health and Epidemiological Surveillance. This domain encompasses the collection, analysis, and dissemination of health-related data for the purpose of preventing and controlling disease within a population. It is inherently cross-disciplinary, drawing on medical epidemiology, data science, geographic information systems, and public policy, which is reflected in the diverse stakeholder group and the mixed technical/non-technical requirement set described below.

1.5 Stakeholders

The following stakeholders were identified through analysis of the problem scenario. Each stakeholder group has distinct goals and concerns that the requirements in this document are designed to satisfy.

- Ministry of Health / Public Health Departments — the primary decision-making body responsible for declaring outbreaks, allocating emergency resources, and setting public health policy based on system outputs.
- Hospitals and Diagnostic Laboratories — data-providing organizations whose case and test records form the primary raw input to the surveillance pipeline.
- Epidemiologists and Data Analysts — technical domain experts who interpret AI predictions, validate hotspot analysis, and refine analytical models.
- Field Health Workers and Community Health Officers — front-line personnel who capture ground-level data in areas with limited digital infrastructure.
- General Public — the ultimate indirect beneficiary of the system through earlier outbreak containment and improved community health outcomes.
- International Health Agencies (e.g., WHO, CDC) — external partners who receive aggregated surveillance reports and may provide comparative global data.
- System Administrators and IT Support Teams — responsible for platform configuration, user management, uptime, and data security.

1.6 Existing Challenges

The problem scenario highlights several systemic weaknesses in current disease-surveillance practice that this system is designed to resolve:

- Delayed Detection: Manual, paper-based, or periodic batch reporting from hospitals means that outbreak signals are often visible only after a disease has already spread significantly within a community.
- Fragmented Reporting: Hospitals, laboratories, and public health agencies frequently operate independent, non-interoperable information systems, making it difficult to build a unified, real-time picture of disease activity.
- Inadequate Predictive Capability: Existing systems are largely descriptive (reporting what has already happened) rather than predictive (forecasting what is likely to happen next), limiting the ability of health authorities to act pre-emptively.
- Absence of Spatial Intelligence: Without GIS-based visualization, it is difficult for officials to quickly identify which geographic areas require urgent intervention.
- Inefficient Resource Allocation: Hospital beds, medical staff, vaccines, and supplies are often allocated reactively rather than being planned ahead of anticipated demand.

1.7 System Objectives

The following objectives, drawn directly from the problem statement, define the measurable outcomes the system must achieve. Each objective is elaborated below to clarify its intended scope.

1. Collect real-time public health data — Continuously ingest structured and unstructured data from hospitals, laboratories, IoT devices, and field workers so that the analytical layer always operates on the most current available information.
2. Predict disease outbreaks using AI — Apply machine learning models trained on historical and real-time data to forecast the likelihood, timing, and severity of emerging outbreaks before they are clinically obvious.
3. Identify outbreak hotspots through GIS — Translate case data into geographic visualizations that reveal spatial clusters and emerging hotspots at a glance.

4. Generate early warning alerts — Automatically notify relevant public health personnel through multiple channels once outbreak-risk indicators cross defined thresholds.
5. Monitor disease transmission trends — Provide continuously updated trend analytics so that analysts can track whether an outbreak is accelerating, plateauing, or declining.
6. Support healthcare resource planning — Translate outbreak forecasts into actionable resource recommendations (beds, staff, medical supplies) for hospitals and health departments.
7. Provide epidemiological analytics dashboards — Present all of the above information through role-appropriate, easy-to-interpret dashboards.
8. Improve public health decision-making — Ultimately support faster, evidence-based policy and operational decisions that reduce disease transmission and protect community health.

1.8 References

- IEEE Std 830-1998 — Recommended Practice for Software Requirements Specifications.
- WHO Guidelines on Epidemic and Pandemic-Prone Disease Surveillance.
- HL7 FHIR Specification — Standard for Healthcare Data Exchange.
- Course material: CSA10 – Software Engineering, Scenario-Based Requirement Analysis, SIMATS Engineering.

2. Overall Description

2.1 Problem Description

Disease outbreaks spread rapidly due to delayed detection, fragmented healthcare reporting systems, and inadequate epidemiological analysis. Existing surveillance systems often lack real-time monitoring and predictive capabilities. Health data collected from hospitals and laboratories is typically siloed, reported manually, and analyzed only after significant time has elapsed — by which point transmission may already be widespread within the affected community.

This delay has direct consequences: hospitals may be caught unprepared for a surge in patients, public health officials may be unable to issue timely public warnings, and scarce medical resources such as vaccines, isolation beds, and trained staff may be misallocated. Furthermore, without a geographic view of case distribution, it is difficult for decision-makers to prioritize interventions in the areas of greatest need.

An intelligent disease surveillance platform is therefore required to collect healthcare data from hospitals, laboratories, and public health agencies in real time; analyze disease trends using AI algorithms; identify potential outbreak hotspots geographically; and provide early warning notifications to decision-makers. The system must also support data visualization, healthcare resource planning, and policy decision-making, in order to minimize disease transmission and improve overall community health outcomes.

2.2 Product Perspective / System Context

The proposed system is a new, standalone cloud-hosted platform that integrates with existing external systems rather than replacing them. It sits between primary data sources (hospitals, laboratories, IoT devices, field workers) and the consumers of surveillance intelligence (public health officials, epidemiologists, and external health agencies). It does not replace hospital electronic health record (EHR) systems, laboratory information systems (LIS), or statutory disease-notification legal processes; instead, it consumes data from these systems and produces derived analytical intelligence.

2.3 Scope of the System

The system will function as a centralized, cloud-hosted public health intelligence platform that performs the following high-level functions:

- Ingests structured and unstructured health data from hospitals, diagnostic laboratories, IoT-based sensors (e.g., wearable devices, environmental sensors), and manual field reports.
- Applies AI/Machine Learning models to detect anomalies and predict the likelihood, location, and spread of disease outbreaks.
- Uses GIS mapping to visualize disease density and identify emerging hotspots.
- Sends automated early-warning alerts (SMS/email/dashboard) to public health officials and relevant agencies.
- Provides dashboards for trend monitoring, resource planning, and reporting to support policy decisions.

Out of Scope

The following items are explicitly excluded from the scope of this system: direct clinical diagnosis of individual patients; replacement of statutory disease-reporting legal frameworks; hospital billing systems; and treatment or prescription record management beyond what is strictly required for surveillance analytics.

2.4 User Classes and Characteristics

User Class	Characteristics
Public Health Officer	Moderate technical proficiency; primarily consumes dashboards and alerts; needs concise, actionable visual summaries rather than raw data.
Epidemiologist / Data Analyst	High technical and statistical proficiency; requires access to raw and processed data, model parameters, and trend detail.
Hospital & Laboratory Staff	Variable technical proficiency; interacts with the system primarily through existing HIS/LIS integrations rather than a dedicated interface.
Field Health Worker	Basic to moderate technical proficiency; uses a simplified mobile interface, often under time pressure and limited connectivity.
System Administrator	High technical proficiency; manages configuration, users, and system health rather than epidemiological content.

2.5 Operating Environment

- Server-side: Cloud infrastructure (e.g., AWS/Azure/GCP-class environment) supporting containerized microservices and managed AI/ML compute.
- Client-side: Modern web browsers (Chrome, Edge, Firefox, Safari) for the dashboard; Android and iOS for the field-worker mobile application.
- Network: Broadband/internet connectivity at hospitals and data centers; intermittent mobile network connectivity tolerated at field locations.

2.6 Design and Implementation Constraints

Design and implementation are constrained by the need to integrate with heterogeneous legacy hospital systems, comply with health-data privacy regulations, and operate within the cloud budget approved by the sponsoring health agency. These constraints are elaborated further in Section 10.

2.7 Assumptions and Dependencies

The system's design assumes the continued availability of external services such as GIS mapping providers, SMS/email gateways, and sufficient historical epidemiological data for model training. These assumptions and their associated risks are elaborated further in Section 10 and Section 12.

3. System Architecture Overview

The system is designed using a layered, microservice-oriented architecture to satisfy the maintainability and scalability requirements described in Section 5. Each layer has a distinct responsibility and communicates with adjacent layers through well-defined APIs, allowing individual components (for example, the AI prediction engine) to be updated or scaled independently without affecting the rest of the platform.

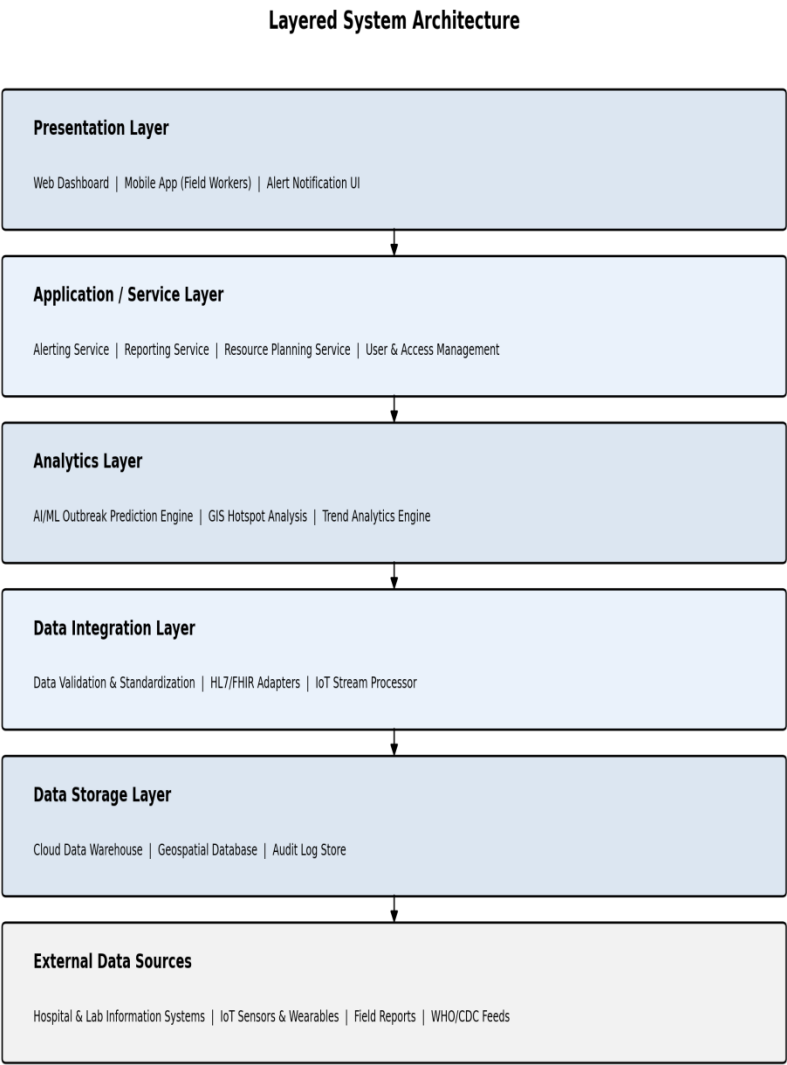


Figure 1: Layered System Architecture

External Data Sources

This layer represents systems and devices outside the platform's direct control: hospital and laboratory information systems, IoT sensors and wearables, manually submitted field reports, and data feeds shared by external health agencies.

Data Integration Layer

Incoming data from all sources is validated, de-duplicated, standardized (including translation into HL7/FHIR-compatible structures where necessary), and queued for storage and analysis. This layer isolates the rest of the system from the inconsistencies typical of real-world healthcare data sources.

Data Storage Layer

Processed data is persisted in a cloud data warehouse for analytical workloads, a geospatial database for GIS operations, and a separate, append-only audit log store to satisfy the auditability requirement described in Section 5.

Analytics Layer

This layer contains the AI/ML outbreak prediction engine, the GIS hotspot analysis engine, and the trend analytics engine. It reads from the data storage layer and produces derived intelligence — risk scores, hotspot boundaries, and trend indicators — that is consumed by the application layer above it.

Application / Service Layer

This layer implements the business logic exposed to end users: the alerting service, reporting service, resource-planning service, and user and access management service. Each service is independently deployable, which supports the maintainability non-functional requirement.

Presentation Layer

The topmost layer consists of the web-based dashboard used by public health officers and epidemiologists, the mobile application used by field health workers, and the notification surfaces used to deliver alerts.

4. Functional Requirements

The functional requirements below define the specific, verifiable behaviours the system must exhibit. They are grouped by module, and each requirement is accompanied by a short rationale explaining why it is necessary, to aid design and test-case derivation.

4.1 Data Collection Module

This module is responsible for acquiring raw health data from all external sources described in Section 3 and preparing it for downstream analysis.

- **FR1:** The system shall collect real-time patient case data (symptoms, diagnosis, lab results) from hospital and laboratory information systems via secure APIs.
***Rationale:** Real-time ingestion is the foundation for early detection; without it, all downstream prediction and alerting would operate on stale data.*
- **FR2:** The system shall ingest data streams from IoT devices such as environmental sensors and wearable health monitors.
***Rationale:** Environmental and physiological signals (e.g., temperature clusters, air quality) can provide early indicators of disease activity before clinical case data is reported.*
- **FR3:** The system shall allow field health workers to manually submit surveillance data through a mobile/web interface, including in low-connectivity, offline-capable mode.
***Rationale:** Many surveillance-relevant regions have unreliable connectivity; offline capture with later synchronization ensures data is not lost.*
- **FR4:** The system shall validate, de-duplicate, and standardize incoming data before storage.
***Rationale:** Multiple overlapping data sources (hospital + lab + field) can report the same case; validation prevents inflated case counts from skewing predictions.*

4.2 AI-Based Outbreak Prediction Module

This module applies machine learning techniques to historical and real-time data in order to forecast the likelihood and severity of disease outbreaks.

- **FR5:** The system shall analyze historical and real-time disease data using AI/ML models to predict potential outbreaks.
Rationale: This directly satisfies Objective 2 (predict disease outbreaks using AI) from Section 1.7.
- **FR6:** The system shall compute an outbreak-risk score for each monitored region and disease category.
Rationale: A quantified risk score allows thresholds to be defined objectively for alerting (see FR11) rather than relying on subjective judgement.
- **FR7:** The system shall allow epidemiologists to retrain or fine-tune predictive models with updated datasets.
Rationale: Disease dynamics change over time (e.g., new variants); models must be able to be updated without a full system redeployment.

4.3 GIS Hotspot Identification Module

This module converts case data into geographic visualizations to support Objective 3 (identify outbreak hotspots through GIS).

- **FR8:** The system shall map reported disease cases geographically using GIS visualization.
Rationale: Spatial visualization allows officials to see patterns not evident from tabular data alone.
- **FR9:** The system shall automatically identify and highlight outbreak hotspots based on case density and growth rate.
Rationale: Automated hotspot detection reduces reliance on manual map interpretation and speeds up response time.
- **FR10:** The system shall allow users to filter hotspot maps by disease type, time period, and administrative region.
Rationale: Filtering supports both broad situational awareness and focused investigation of a specific disease or jurisdiction.

4.4 Early Warning & Alerting Module

This module notifies relevant stakeholders once analytical outputs indicate an elevated risk of outbreak, satisfying Objective 4.

- **FR11:** The system shall generate automated early-warning alerts when predefined outbreak-risk thresholds are exceeded.
Rationale: Automated thresholding ensures alerts are issued consistently and without the delay of manual review.
- **FR12:** The system shall deliver alerts via SMS, email, and in-app dashboard notifications to designated stakeholders.
Rationale: Multiple channels increase the likelihood that a time-sensitive alert reaches the responsible official promptly.
- **FR13:** The system shall allow public health officers to acknowledge, escalate, or dismiss generated alerts.
Rationale: A closed-loop acknowledgment workflow ensures accountability and provides an audit trail of the response to each alert.

4.5 Trend Monitoring & Analytics Module

This module supports Objective 5 by providing continuously updated views of how disease transmission is evolving.

- **FR14:** The system shall track and display disease transmission trends over configurable time periods.
Rationale: Configurable time windows allow analysts to examine both short-term spikes and long-term seasonal patterns.

- **FR15:** The system shall compare current trends against historical baselines and seasonal patterns.
Rationale: Comparing against a baseline helps distinguish a genuine emerging outbreak from normal seasonal variation.

4.6 Resource Planning Module

This module translates predictive intelligence into actionable operational recommendations, satisfying Objective 6.

- **FR16:** The system shall recommend healthcare resource allocation (beds, staff, vaccines, medical supplies) based on predicted case loads.
Rationale: Proactive, forecast-driven allocation reduces the risk of hospitals being overwhelmed during an outbreak surge.
- **FR17:** The system shall allow administrators to record and update available healthcare resource inventories.
Rationale: Accurate current-inventory data is required as an input to any meaningful resource recommendation.

4.7 Dashboard & Reporting Module

This module presents analytical outputs to end users and supports external reporting, satisfying Objectives 7 and 8.

- **FR18:** The system shall provide role-based epidemiological dashboards with visual analytics (charts, maps, trend lines).
Rationale: Role-based views ensure each stakeholder sees information relevant to their responsibilities without being overwhelmed by irrelevant detail.
- **FR19:** The system shall generate exportable reports (PDF/Excel) summarizing outbreak status and forecasts.
Rationale: Exportable reports are required for offline briefings, government meetings, and record-keeping.
- **FR20:** The system shall support secure data sharing with external health agencies (e.g., WHO, CDC).
Rationale: International reporting obligations and collaborative response require standardized, secure data exchange with external bodies.

4.8 User & Access Management Module

This module underpins the security and auditability of the entire platform.

- **FR21:** The system shall provide role-based authentication and authorization for all user categories.
Rationale: Role-based access control ensures users can only view or modify data appropriate to their responsibilities, satisfying the Security non-functional requirement.
- **FR22:** The system administrator shall be able to create, modify, and deactivate user accounts and roles.
Rationale: Centralized account lifecycle management is required for onboarding new facilities and revoking access when staff change roles.

5. Non-Functional Requirements

Non-functional requirements define the quality attributes the system must exhibit, independent of specific features. Each is stated in a measurable form so that it can be objectively verified during testing.

Category	Requirement Description
Performance	The system shall process and analyze incoming health data streams and refresh dashboards within 5 seconds under normal load conditions.
Scalability	The system shall scale horizontally on cloud infrastructure to support onboarding of additional hospitals, regions, and IoT devices without redesign.

Security	All health data shall be encrypted in transit (TLS 1.2+) and at rest (AES-256); access shall be role-based and comply with data protection regulations (e.g., HIPAA/GDPR-equivalent).
Reliability	The system shall maintain accurate outbreak predictions with a documented model confidence score and shall log all data-ingestion failures for review.
Availability	The core surveillance and alerting services shall maintain 99.5% uptime, with redundant cloud deployment across availability zones.
Usability	Dashboards shall be intuitive for non-technical public health officials, requiring no more than 2 hours of training to operate core features.
Maintainability	The system shall follow a modular, microservice-based architecture to allow independent updates to AI models, GIS modules, and alerting services.
Interoperability	The system shall support standard healthcare data exchange formats such as HL7/FHIR for integration with hospital information systems.
Auditability	The system shall maintain an immutable audit log of all data submissions, alert generations, and administrative actions.

5.1 Discussion of Key Quality Attributes

Performance and Availability are treated as the most safety-critical attributes for this system, since a delay or outage during an active outbreak could directly cost lives; both are therefore backed by redundant cloud deployment and a documented, measurable target (5-second refresh, 99.5% uptime) rather than a vague qualitative goal.

Security and Auditability are of particular importance given that the system processes sensitive health data; encryption, role-based access, and immutable logging are treated as mandatory, non-negotiable requirements rather than desirable features.

Usability is prioritized for the Public Health Officer user class in particular, since this group is expected to have limited technical training but must be able to act quickly and confidently on dashboard information during a crisis.

6. Actors and Use Cases

6.1 Primary Actors

Primary actors initiate interactions with the system to accomplish a goal. Each is described below along with its principal responsibilities.

Actor	Description
Public Health Officer	Monitors dashboards, reviews and responds to alerts, and makes policy and resource-allocation decisions based on system recommendations.
Epidemiologist / Data Analyst	Analyzes disease trends in depth, tunes and validates AI prediction models, and investigates GIS-identified hotspots.
Hospital & Laboratory Staff	Submits patient case data and laboratory test results into the system, typically via automated integration with existing HIS/LIS systems.
Field Health Worker	Collects and submits ground-level surveillance data, especially in remote or under-resourced areas lacking digital hospital infrastructure.

System Administrator	Manages user accounts, roles, system configuration, integration settings, and overall data integrity.
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6.2 Secondary Actors

Secondary actors are external systems or services that the platform depends on to fulfil primary actors' goals.

Actor	Description
AI Prediction Engine	Automated component that executes machine learning models to forecast outbreak likelihood, location, and severity.
GIS Mapping Service	External or embedded mapping service used to render geographic hotspot visualizations.
IoT Sensor Network	Connected devices supplying environmental and health-monitoring data streams to the platform.
Alert Gateway (SMS/Email)	Third-party gateway used to dispatch early-warning notifications to stakeholders.
External Health Agencies (WHO/CDC)	Receive shared reports and aggregated surveillance data for cross-border and global health monitoring.

6.3 Use Case Summary (by Actor)

- Public Health Officer → Login/Manage Access, View Epidemiological Dashboard, Acknowledge/Escalate Alerts, Plan Healthcare Resource Allocation, Manage System Users & Config (delegated).
- Epidemiologist / Data Analyst → Analyze Disease Transmission Trends, Predict Disease Outbreak (AI), Identify Outbreak Hotspots (GIS), Generate Reports & Share with Agencies.
- Hospital & Laboratory Staff → Collect Health & Lab Report Data.
- Field Health Worker → Submit Field Surveillance Data.
- System Administrator → Manage System Users & Config.
- AI Prediction Engine → Predict Disease Outbreak (AI).
- GIS Mapping Service → Identify Outbreak Hotspots (GIS).
- IoT Sensor Network → Monitor IoT Sensor Feeds.
- Alert Gateway → Generate Early Warning Alerts.
- External Health Agencies → Generate Reports & Share with Agencies.

6.4 Use Case Diagram

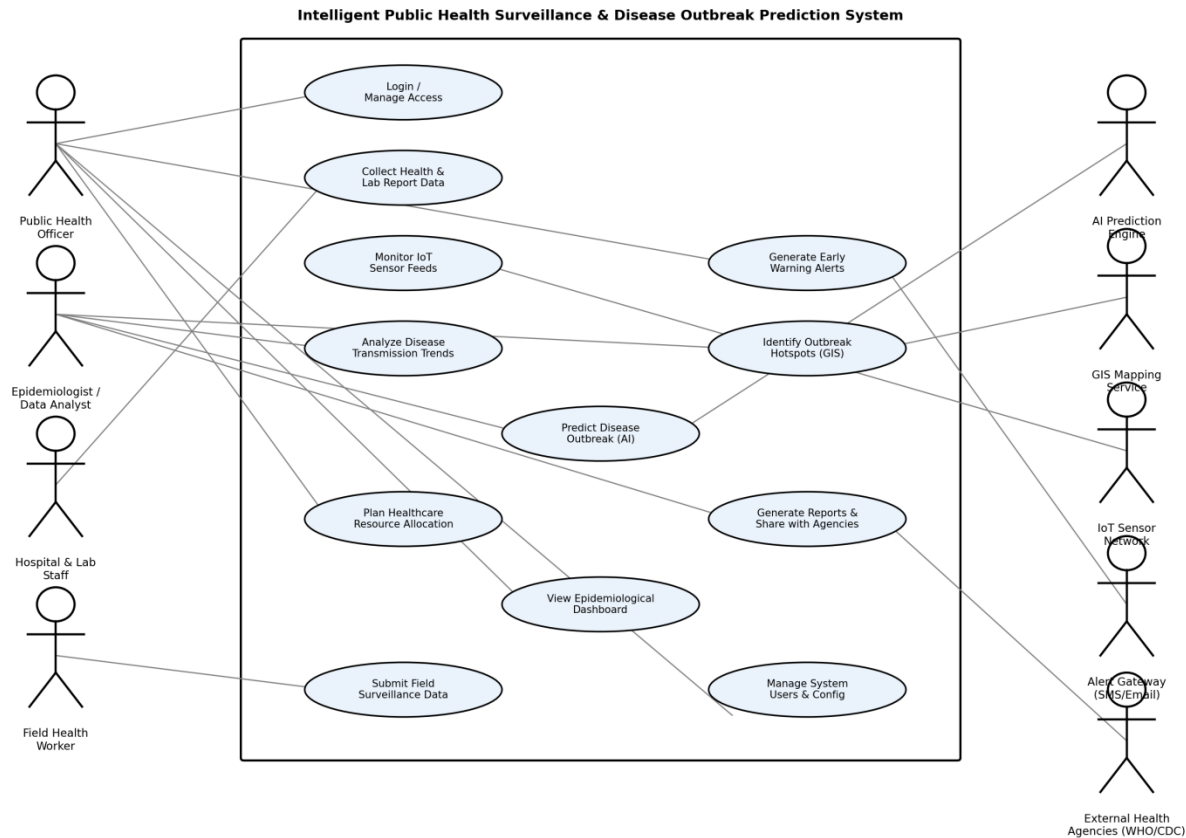


Figure 2: Use Case Diagram — Intelligent Public Health Surveillance and Disease Outbreak Prediction System

6.5 Detailed Use Case Descriptions

The following six use cases have been selected for detailed description as they represent the core value chain of the system, from data intake through to decision-making action.

Use Case 1: Collect Health & Lab Report Data

Use Case	Collect Health & Lab Report Data
Actors	Hospital & Laboratory Staff (primary)
Preconditions	The hospital/laboratory information system is registered and authenticated with the platform's data-integration layer.
Main Flow	1) Staff enters or the HIS/LIS automatically transmits case/lab data. 2) System receives data via secure API. 3) System validates format and completeness. 4) System de-duplicates against existing records. 5) System stores standardized record and timestamps it.
Alternate Flow	If a required field is missing, the system flags the record as incomplete and requests correction rather than silently discarding it.
Postconditions	A validated, standardized case record is available to the analytics layer.
Exceptions	API connection failure → system queues the submission for retry and alerts the administrator if failures persist beyond a configured threshold.

Use Case 2: Submit Field Surveillance Data

Use Case	Submit Field Surveillance Data
Actors	Field Health Worker (primary)
Preconditions	The worker has an authenticated session on the mobile application, with or without active internet connectivity.
Main Flow	1) Worker records observations (symptoms, location, case counts) in the mobile app. 2) If online, data is transmitted immediately. 3) If offline, data is stored locally and queued. 4) Once connectivity is restored, queued data is automatically synchronized to the server.
Postconditions	Field data is merged into the central data store and made available for validation (FR4) and analysis.
Exceptions	Sync conflict (e.g., duplicate submission) → system applies de-duplication logic and retains the most complete record.

Use Case 3: Predict Disease Outbreak (AI)

Use Case	Predict Disease Outbreak (AI)
Actors	Epidemiologist / Data Analyst (primary), AI Prediction Engine (secondary)
Preconditions	Historical and real-time disease data is available and validated in the system.
Main Flow	1) Analyst selects a region and disease category. 2) System runs the AI model on aggregated data. 3) System computes an outbreak-risk score. 4) System displays the prediction with a confidence interval.
Postconditions	The outbreak prediction is stored and made available to the alerting module (FR11).
Exceptions	Insufficient data → system flags a low-confidence prediction and notifies the analyst rather than presenting a misleadingly precise figure.

Use Case 4: Identify Outbreak Hotspots (GIS)

Use Case	Identify Outbreak Hotspots (GIS)
Actors	Epidemiologist / Data Analyst (primary), GIS Mapping Service (secondary)
Preconditions	Geolocated case data is available for the selected time period and region.
Main Flow	1) Analyst opens the GIS hotspot view. 2) System queries the geospatial database for case density. 3) System calculates growth rate per area. 4) System renders a map highlighting emerging hotspots, colour-coded by severity.
Postconditions	Hotspot boundaries are stored and made available to the reporting and alerting modules.
Exceptions	Mapping service unavailable → system falls back to a cached tile layer and displays a data-currency warning to the user.

Use Case 5: Generate Early Warning Alerts

Use Case	Generate Early Warning Alerts
Actors	Public Health Officer (primary), Alert Gateway (secondary)
Preconditions	An outbreak-risk score has exceeded the configured alert threshold.

Main Flow	1) System detects the threshold breach. 2) System composes an alert with region/disease details. 3) System dispatches the alert via SMS/email/dashboard. 4) Officer acknowledges, escalates, or dismisses the alert.
Postconditions	The alert status (acknowledged/escalated/dismissed) is logged for audit.
Exceptions	Gateway delivery failure → system retries delivery and logs the failure for administrator review.

Use Case 6: Plan Healthcare Resource Allocation

Use Case	Plan Healthcare Resource Allocation
Actors	Public Health Officer (primary)
Preconditions	A current outbreak forecast and up-to-date resource inventory data are available.
Main Flow	1) Officer opens the resource-planning view for an affected region. 2) System compares predicted case load against current inventory (beds, staff, supplies). 3) System recommends allocation adjustments. 4) Officer approves, modifies, or rejects the recommendation.
Postconditions	Approved allocation changes are recorded and made available for reporting.
Exceptions	Inventory data is stale (not updated within a configured window) → system warns the officer that the recommendation may be unreliable.

7. Data Requirements

This section defines the categories of data the system must capture, store, and manage, along with key data-quality and lifecycle considerations.

7.1 Data Dictionary (Key Entities)

Data Entity	Description
Case Record	Anonymized demographic attributes, symptoms, diagnosis codes, lab results, reporting facility, and timestamp for an individual reported case.
Laboratory Result	Test type, pathogen identified, result status, and reporting laboratory linked to a case record.
Geospatial Record	Facility and case location coordinates together with administrative boundary identifiers used for GIS analysis.
IoT Sensor Reading	Environmental readings (e.g., temperature, air quality) or wearable vital-sign data with a device identifier and timestamp.
Prediction Record	Outbreak-risk score, confidence interval, disease category, region, and model version that produced it.
Alert Record	Alert content, trigger threshold, delivery channel(s), timestamp, and acknowledgment/escalation status.
Resource Inventory Record	Hospital bed capacity, staff availability, and medical supply stock levels per facility, with a last-updated timestamp.
User & Audit Record	User role, permissions, login history, and a full log of data submissions and administrative actions.

7.2 Data Quality and Privacy

All personally identifiable patient information shall be anonymized or pseudonymized prior to analytics processing, in compliance with applicable health-data privacy regulations. Incoming data is subject to automated validation rules (completeness, format, and range checks) before being accepted into the analytical data store, as described in FR4.

7.3 Data Retention and Lifecycle

Case-level and laboratory data shall be retained in an identifiable form only for the minimum period required by public health regulation, after which records shall be further anonymized for long-term historical trend analysis. Audit log records shall be retained in full, immutable form for the duration required by the organization's compliance policy.

8. External Interface Requirements

8.1 User Interfaces

- Web-based dashboard for public health officers and epidemiologists, optimized for desktop use with support for common screen resolutions.
- Mobile application (Android/iOS) for field health workers, supporting offline data capture with later synchronization, and a simplified interface appropriate for low-connectivity, time-constrained field use.

8.2 Hardware Interfaces

- Integration with IoT sensors and wearable devices via standard wireless protocols (e.g., MQTT, Bluetooth Low Energy) for environmental and physiological data collection.

8.3 Software Interfaces

- REST/HL7-FHIR APIs for integration with Hospital Information Systems (HIS) and Laboratory Information Systems (LIS).
- GIS mapping service API for spatial visualization and hotspot rendering.
- Cloud storage and compute services for data warehousing and AI model training/execution.

8.4 Communication Interfaces

- SMS and email gateway APIs for alert dispatch to public health officials and other stakeholders.
- Secure data-exchange interface (e.g., encrypted API or SFTP) for reporting to external health agencies such as WHO or CDC.

9. System Features

This section summarizes the system's core features from a user-facing perspective, consolidating the functional requirements described in Section 4 into feature-level statements suitable for a project overview or stakeholder presentation.

Feature	Priority	Summary
Real-time Data Ingestion	High	Continuous, validated intake of case, lab, IoT, and field data from all connected sources.

AI-Driven Outbreak Prediction	High	Predictive modeling with a quantified, region-specific outbreak-risk score and confidence level.
GIS Hotspot Visualization	High	Interactive, filterable maps highlighting emerging outbreak hotspots by disease and region.
Automated Early-Warning Alerts	High	Multi-channel alerting with threshold-based triggers and an acknowledgment workflow.
Trend Monitoring Dashboards	Medium	Configurable views comparing current transmission trends against historical baselines.
Resource Planning Recommendations	Medium	Forecast-driven suggestions for hospital bed, staff, and supply allocation.
Reporting & External Data Sharing	Medium	Exportable reports and secure data exchange with agencies such as WHO/CDC.
Role-Based Access & Audit Logging	High	Secure, role-appropriate access control with a complete, immutable audit trail.

10. Assumptions and Constraints

10.1 Assumptions

The following assumptions were made during requirement analysis. Each carries an associated risk if it does not hold true, which is addressed further in Section 12 (Risk Analysis).

- Participating hospitals and laboratories have digital information systems capable of API-based integration; facilities without such systems will require a manual or semi-automated onboarding process.
- Reasonably reliable internet connectivity is available at most reporting facilities; field workers may occasionally operate offline, which is accommodated by FR3.
- Sufficient historical disease data exists to train and validate AI prediction models; regions with very limited historical data may initially receive lower-confidence predictions.
- Public health authorities will define and formally approve outbreak-risk alert thresholds prior to go-live, rather than the system choosing thresholds unilaterally.

10.2 Constraints

- Technical: The system must integrate with heterogeneous legacy hospital IT systems of varying data quality and API maturity.
- Operational: Field data collection depends on health-worker availability, training, and device access in remote regions.
- Economic: Cloud infrastructure and AI compute costs must remain within the budget allocated by the sponsoring health agency.
- Legal/Regulatory: The system must comply with national and international health-data privacy laws (e.g., HIPAA, GDPR-equivalent frameworks) and any statutory disease-notification requirements.
- Time: The initial phase must be deployable within the project timeline defined by the sponsoring health agency, which may limit the scope of the first release.

11. Requirement Traceability Matrix

The traceability matrix below maps each system objective from Section 1.7 to the functional requirements that implement it, allowing reviewers to confirm that every objective is addressed and that no functional requirement is orphaned from a business goal.

Objective	Related Functional Requirements
1. Collect real-time public health data	FR1, FR2, FR3, FR4
2. Predict disease outbreaks using AI	FR5, FR6, FR7
3. Identify outbreak hotspots through GIS	FR8, FR9, FR10
4. Generate early warning alerts	FR11, FR12, FR13
5. Monitor disease transmission trends	FR14, FR15
6. Support healthcare resource planning	FR16, FR17
7. Provide epidemiological analytics dashboards	FR18, FR19, FR20
8. Improve public health decision-making	FR18, FR19, FR21, FR22 (supported indirectly by all modules)

12. Risk Analysis

The following risks were identified during requirement analysis, together with their likelihood, potential impact, and proposed mitigation. This analysis directly informs several of the assumptions and constraints in Section 10.

Risk	Likelihood	Impact	Mitigation
Legacy hospital systems cannot integrate via standard APIs.	Medium	High	Provide a manual/CSV upload fallback and a phased onboarding plan for lower-maturity facilities.
Insufficient historical data for AI model training in some regions.	Medium	Medium	Use transfer learning from comparable regions and clearly flag low-confidence predictions to users.
Alert fatigue from excessive or poorly tuned notifications.	Medium	Medium	Allow thresholds to be configured and reviewed by epidemiologists; support alert acknowledgment/escalation tracking.
Data privacy breach involving sensitive health data.	Low	High	Enforce encryption in transit and at rest, strict role-based access, and regular third-party security audits.
Field connectivity issues cause data loss.	Medium	Medium	Offline-capable mobile data capture with automatic synchronization once connectivity is restored (FR3).
Third-party GIS or alert-gateway service outage.	Low	Medium	Implement cached fallback map tiles and multi-channel alert delivery so no single external dependency is a point of total failure.

13. Validation of Requirement Completeness and Consistency

13.1 Stakeholder Coverage Check

All identified stakeholders — public health officers, epidemiologists, hospital/lab staff, field workers, administrators, and external agencies — have corresponding functional requirements and use cases, confirming complete stakeholder coverage. This was verified by cross-referencing the stakeholder list in Section 1.5 against the actor list in Section 6.1–6.2 and the traceability matrix in Section 11.

13.2 Ambiguity, Redundancy, and Missing Requirement Check

- **Ambiguity:** The term "real-time" in the original problem scenario has been quantified in the Non-Functional Requirements (5-second dashboard refresh) to remove ambiguity.
- **Redundancy:** Data collection requirements (FR1–FR3) were reviewed to ensure each source (hospital, IoT, field) is uniquely scoped without functional overlap.
- **Missing Requirements Identified & Added:** Offline data capture for field workers (FR3), audit logging (Auditability non-functional requirement), and HL7/FHIR interoperability (Interoperability non-functional requirement) were added during review, as the original scenario did not explicitly mention them but they are necessary for a workable system.
- **Consistency:** Terminology (e.g., "outbreak-risk score", "hotspot") is used consistently across the Functional Requirements, Use Case Descriptions, and Data Dictionary to avoid confusion during implementation.

13.3 Suggested Improvements

- Define explicit Service Level Agreements (SLAs) with each integrated hospital/laboratory for data submission frequency and format.
- Introduce a feedback loop where confirmed outbreak outcomes are fed back into the AI model to improve prediction accuracy over time.
- Add a multilingual interface for field health workers operating across linguistically diverse regions.
- Conduct periodic third-party security audits given the sensitivity of the health data being processed.
- Pilot the system in a single region before full-scale rollout, to validate model accuracy and alert-threshold calibration under real conditions.

14. Glossary of Terms

Term	Definition
Outbreak	A sudden increase in the number of cases of a disease above what is normally expected in a defined community or region.
Hotspot	A geographic area exhibiting an unusually high density or growth rate of reported disease cases.
Outbreak-Risk Score	A quantified, model-generated indicator of the likelihood and severity of an outbreak for a given region and disease category.
HL7/FHIR	Health Level Seven / Fast Healthcare Interoperability Resources — a standard for exchanging healthcare information electronically.
GIS	Geographic Information System — a framework for capturing, storing, and analyzing spatial/geographic data.
IoT	Internet of Things — network-connected physical devices (e.g., sensors, wearables) that collect and transmit data.
SRS	Software Requirement Specification — a document describing the intended behaviour and constraints of a software system.

Role-Based Access Control (RBAC)	A security approach that restricts system access based on the role assigned to a user.
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15. Conclusion

This Software Requirement Specification has translated the scenario for an Intelligent Public Health Surveillance and Disease Outbreak Prediction System into a structured, verifiable set of functional and non-functional requirements, actors, use cases, and supporting analysis. The document establishes twenty-two functional requirements across eight modules, nine non-functional quality attributes, six primary and five secondary actors, and a traceability matrix confirming that every stated system objective is fully addressed by at least one functional requirement.

The architecture overview, data requirements, and external interface requirements provide sufficient technical grounding for a development team to begin high-level design, while the risk analysis and validation sections surface open issues — such as legacy-system integration and alert-threshold calibration — that should be resolved collaboratively with the sponsoring public health agency before implementation begins.

Taken together, this SRS is intended to serve as a stable, agreed-upon baseline against which the system can be designed, built, tested, and ultimately evaluated for its ability to enable earlier outbreak detection, more efficient healthcare resource allocation, and improved community health outcomes.